

Class 5: Data Viz with ggplot2

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Background

There are many graph systems in R for making plots and figures. These include so-called “*base R*” graphics like the ‘plot()’ function and add on packages like **ggplot2**.

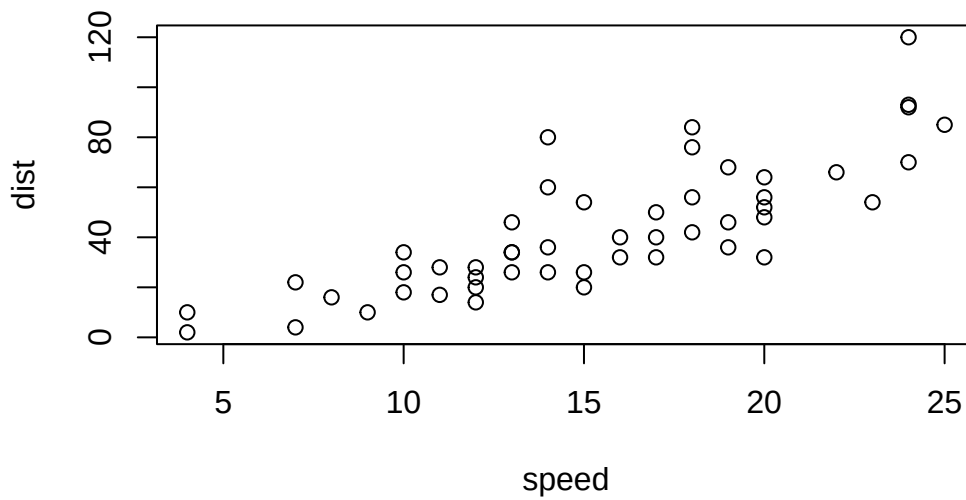
Let’s compare how we make a simple figure with these two systems:

We can use the in-built ‘cars’ dataset:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```



Before I can use ggplot2 I need to install it on my computer. To do this we can use the function 'install.packages("ggplot2")

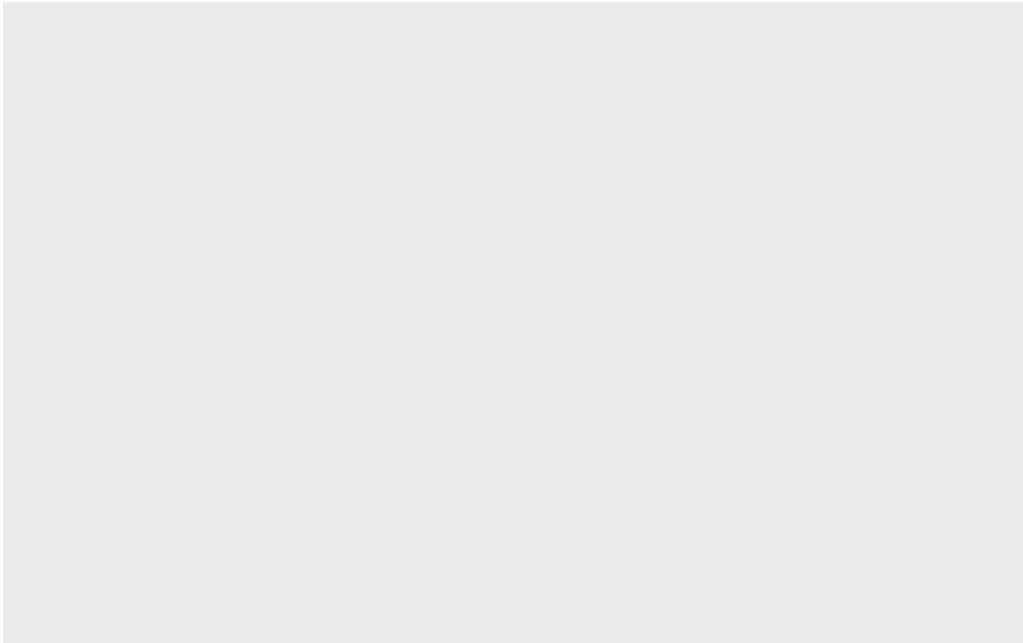
N.B. We never run 'install.packages()' in our quarto doc (we run it once only in our R console) as it would re-install the package every time we render our quarto report.

Once installed we need to load up the package into our R brain:

```
library(ggplot2)
```

The main function in the **ggplot2** package is called 'ggplot()'

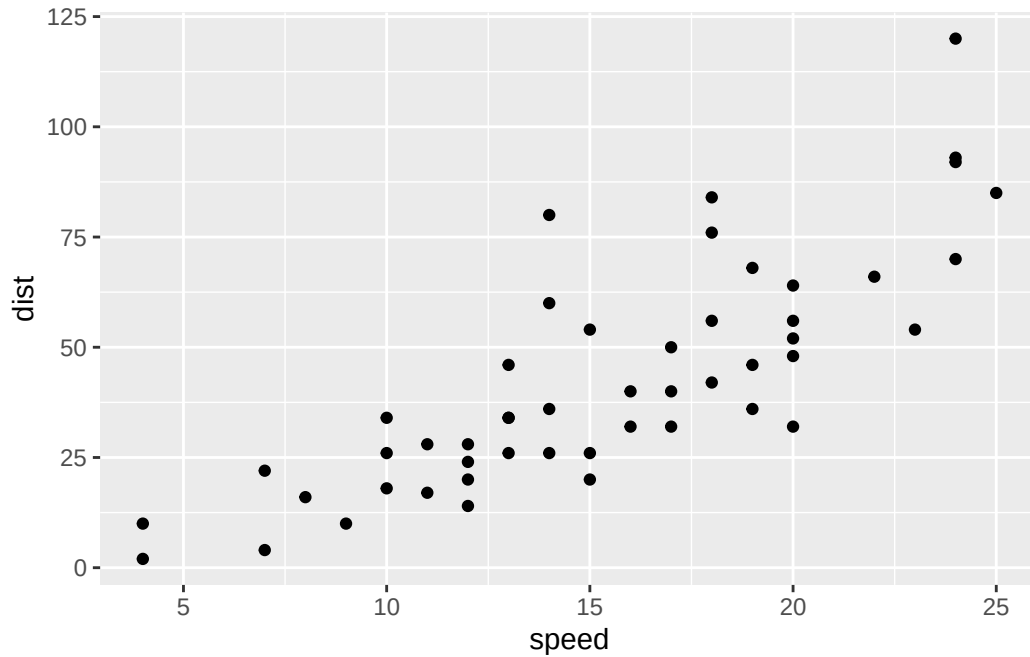
```
ggplot(cars)
```



Every ggplot has at least 3 layers:

- the **data** (a data.frame of the stuff we want to plot)
- the ***aes***thetics (how the data maps to the plot)
- the **geom** layer (how you want the plot drawn, e.g. points, lines, etc.)

```
ggplot(cars) +  
  aes(x = speed, y = dist) +  
  geom_point()
```

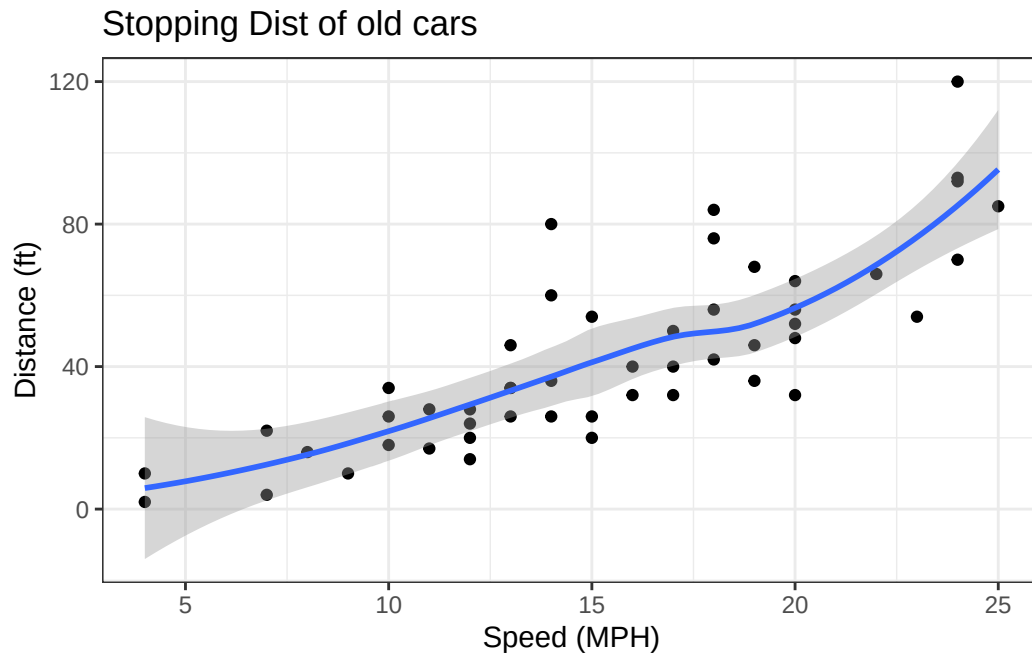


Add some custom features

Let's add a trend line that shows the relationship between speed and distance.

```
ggplot(cars) +  
  aes(x = speed, y = dist) +  
  geom_point() +  
  geom_smooth() +  
  theme_bw() +  
  labs(title= "Stopping Dist of old cars",  
        x = "Speed (MPH)",  
        y = "Distance (ft)")
```

``geom_smooth()`` using `method = 'loess'` and `formula = 'y ~ x'`



Q. Can you make the 'geom_smooth()' function produce a linear straight line fit to the data and turn off the "gray" error region.

Gene Expression

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

```
sum(genes$State == "up")
```

```
[1] 127
```

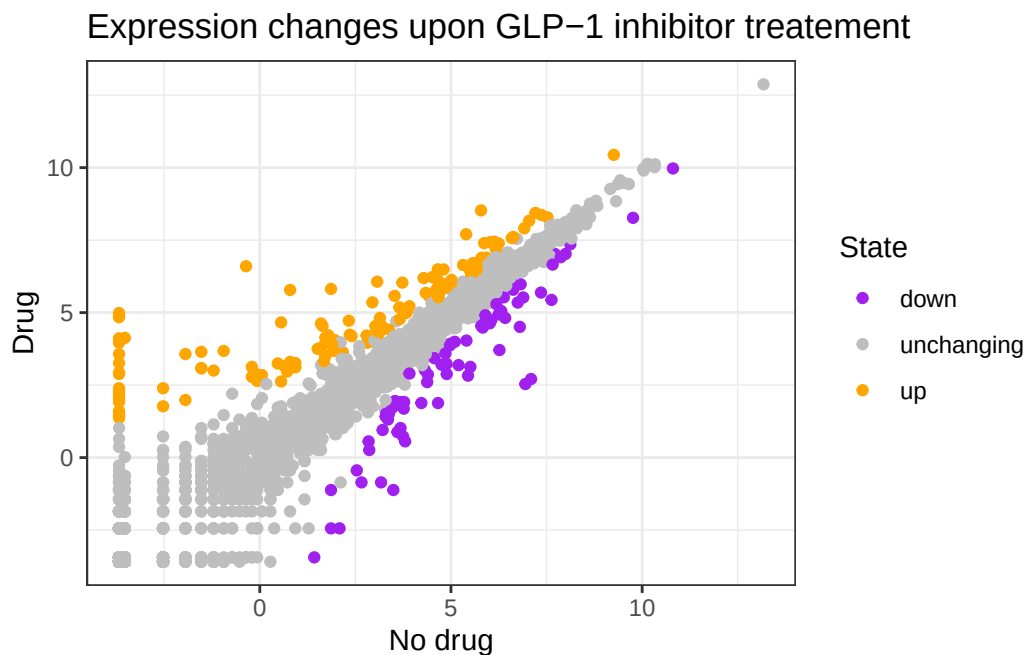
A useful new function in this context is the 'table()' function:

```
table(genes$State)
```

down	unchanging	up
72	4997	127

My first plot attempt

```
ggplot(genes) +  
  aes(Condition1, Condition2, col = State) +  
  geom_point() +  
  scale_colour_manual(values=c("purple","gray","orange")) +  
  theme_bw() +  
  labs(x= "No drug", y = "Drug",  
        title = "Expression changes upon GLP-1 inhibitor treatement")
```



Going further

Here we read the famous gapminder dataset

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"

gapminder <- read.delim(url)
head(gapminder)
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	1952	28.801	8425333	779.4453
2	Afghanistan	Asia	1957	30.332	9240934	820.8530
3	Afghanistan	Asia	1962	31.997	10267083	853.1007
4	Afghanistan	Asia	1967	34.020	11537966	836.1971
5	Afghanistan	Asia	1972	36.088	13079460	739.9811
6	Afghanistan	Asia	1977	38.438	14880372	786.1134

How many entries (i.e. rows) are in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

Q. How many diff “country” entries are in this dataset?

```
length(table(gapminder$country))
```

```
[1] 142
```

```
length(unique(gapminder$country))
```

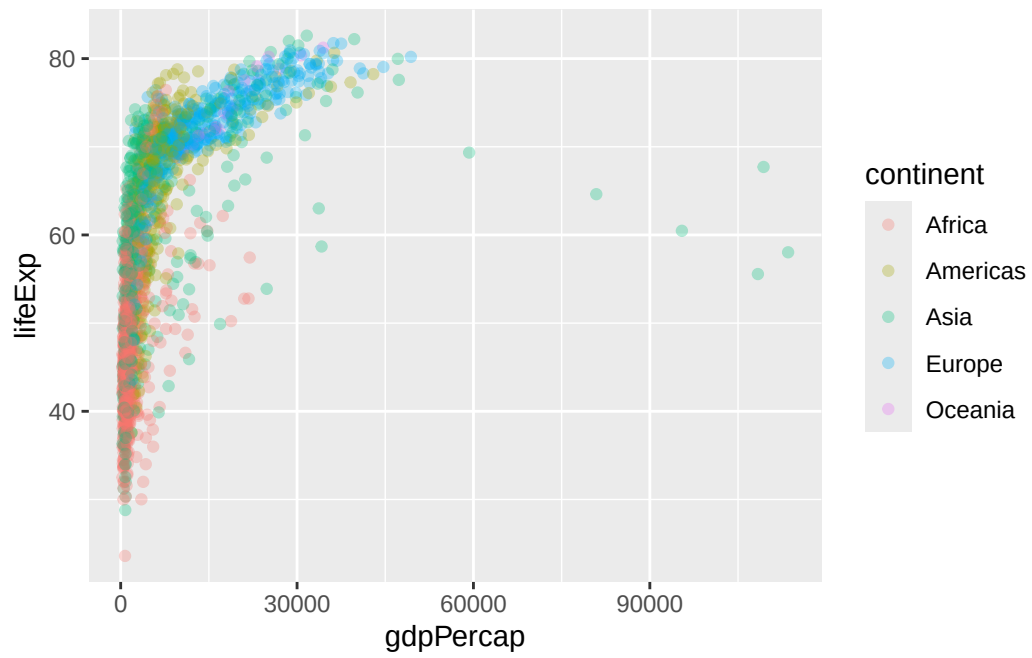
```
[1] 142
```

Let’s make our first plot of the entire dataset.

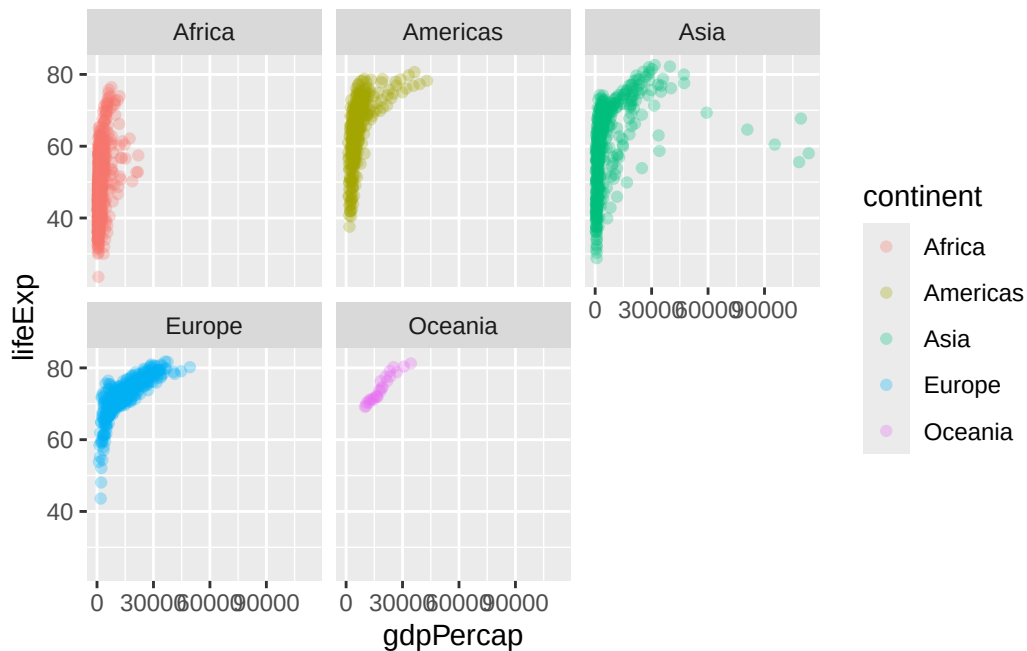
Plot of “gdpPercap” vs “lifeExp” colored by “continent”

```
p <- ggplot(gapminder) +
  aes(gdpPercap, lifeExp, color = continent) +
  geom_point(alpha = 0.3)
```

```
p
```



```
p +  
  facet_wrap(~continent)
```



Make a plot for years 1977 and 2007 only (not all the years in the dataset).

Q. First use the **dplyr** package and the 'filter()' function from that package to extract the year 2007.

```
library(dplyr)
```

```
g07 <- filter(gapminder, year == 2007)
g77 <- filter(gapminder, year == 1977)
g <- filter(gapminder, year == 2007 | year == 1977)
```

```
ggplot(g) +
  aes(x=gdpPerCap, y=lifeExp, color=continent, size=pop) +
  geom_point() +
  facet_wrap(~year)
```

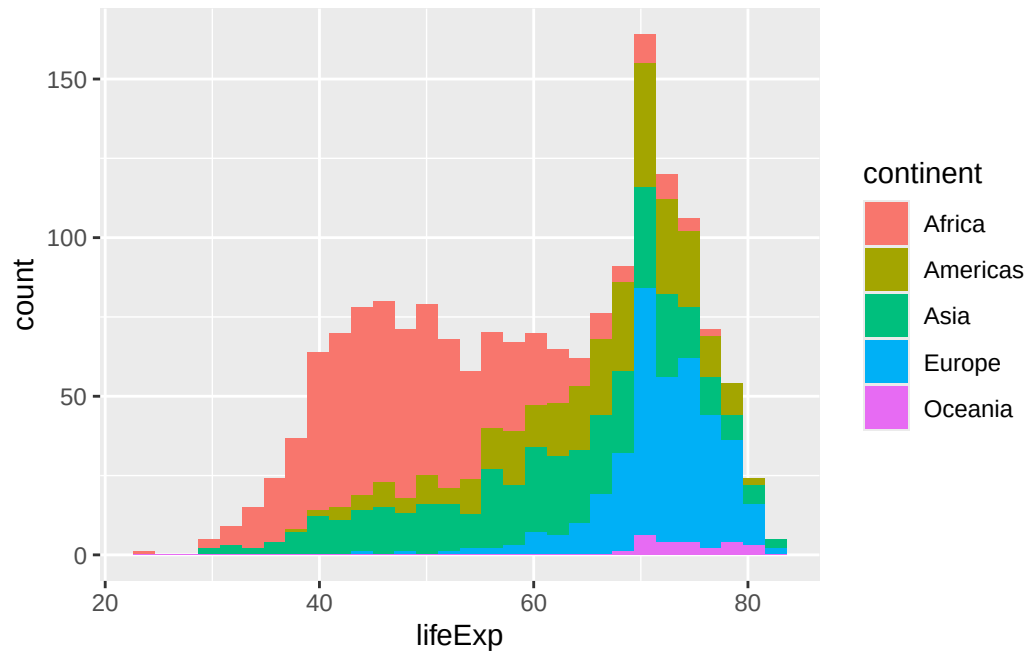


Q. Make a histogram of lifeExp colored by continent (try using 'fill=continent' or 'col=continent')

Q. Make a histogram of lifeExp faceted by continent

```
ggplot(gapminder) +
  aes(x = lifeExp, fill = continent) +
  geom_histogram()
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
ggplot(gapminder) +  
  aes(x = lifeExp) +  
  geom_histogram() +  
  facet_wrap(~continent)
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

